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Royal College - Colombo 07

Grade 12

First Term test – June 2023

Physics I

01 E I

Time : 1 hour

$g = 10\ N\ kg^{-1}$

Answer all questions.

(1) The dimensions of change of momentum is

1) MLT^{-2} 2) MLT^{-1} 3) $ML^{-1}T^{-1}$ 4) $ML^{-1}T^{-2}$ 5) No diamensions

(2) The physical quantity and it's unit is correctly represented by,

1) momentum – Ns^{-1} 2) Impulse – $kg\ ms^{-1}$ 3) density – $g\ cm^{-3}$

4) pressure – $kgm^{-1}\ s^{-2}$ 5) luminance intensity – rad

(3) The ratio of the prinfixes is equal to 10^{-3} is given by,

1)	2)	3)	4)	5)
milli / kilo	killo / micro	Giga / Mega	Giga / Tera	nano / milli

(4) Consider the following statements regarding the dimensions.

A) The physical equations which are existed with logarithmic terms can be constructed using the dimensions.

B) The physical equations which are existed with trignometric ratioes can be constructed using the dimensions.

C) The physical equations which are related with four physical quantities can be constructed using the dimensions.

The true statement(s) from above is/are

1) A only 2) B only 3) C only

4) A and B only 5) B and C only

(5) A vector, scalar and a diamension less physical quantities are represented in order by,

1) momentum, force, relative density 2) displacement, time, frequency

3) pressure, distance, relative velocity 4) acceleration, energy, relative velocity

5) velocity, pressure, relative density

(6) The wrong statement regarding the meter ruler is,

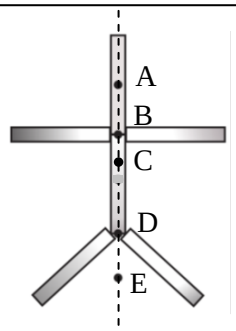
1) The maximum absolute error that can be occured when taking reading using a meter ruler is 1 mm.

2) Meter ruler is divided in 1 mm divisions.

3) Meter ruler is calibrated in centimeters.

4) The minimum measurement that can be taken by a meter ruler as a laboratory measurement is 10 cm.

5) The no parallelax error has to be considered when taking a reading using a meter ruler.

(7) 

The given frame is assembled by six identical rods. The centre of gravity of the frame is most possible at the point OF,

1) A 2) B 3) C
4) D 5) E

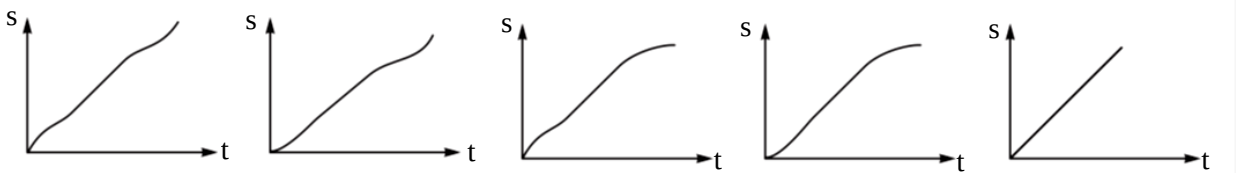
(8) Consider the following statements regarding the apparatus of parellelogram of force in the school lab.

A) The pullies are not on a same plane.
B) The pullies are not at a same horizontal level
C) The pans are connected at the end of the strings.

The reason(s) for not been vertical the diagonal of the parallelogram which is constructed to verify the law of parallelogram of forces is/are

1) A only 2) B only 3) C only
4) A and B only 5) A and C only

(9) The displacement – time graph of a lift which starts to move from the ground floor and ends at the third floor is best represented by,

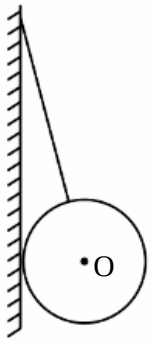


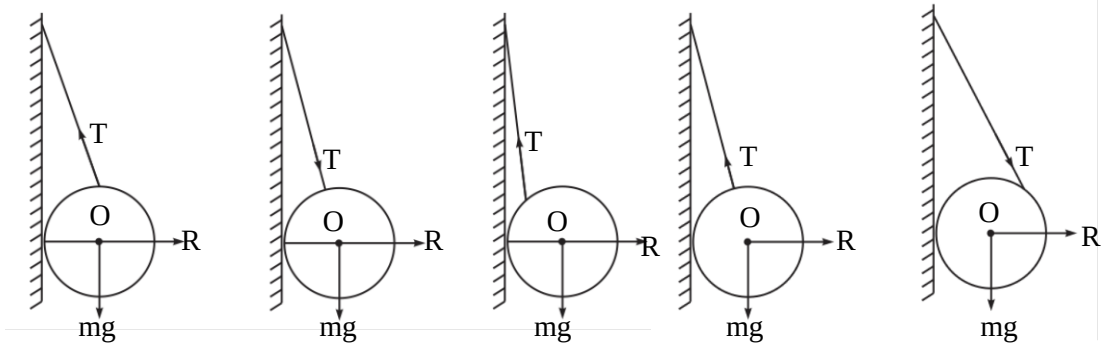
(1) (2) (3) (4) (5)

(10) The addition of two vectors **M** and **N** is equal to **P**. **M** and **N** are inclined in 135° <u>each other.</u> **P** and **M** are perpendicular to each other. The magnitude of **M** is 13. The magnitude of **P** is given by,

1) 13 2) $\frac{13}{\sqrt{2}}$ 3) $13\sqrt{2}$ 4) $\frac{13\sqrt{3}}{2}$ 5) $\frac{13}{2}$

(11) A spherical object is hung on a wall with an inextensible string which is connected to a point at the circumference of the sphere as shown in the figure. The correct free body diagram is given by





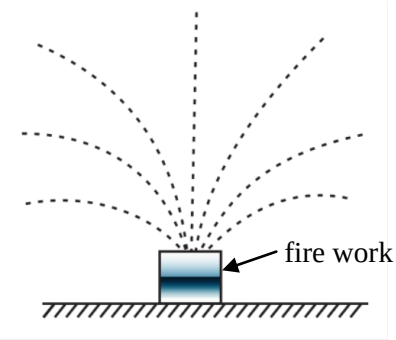
(1) (2) (3) (4) (5)

(12) The percentage errors when measuring the length of a side (l) and the mass (m) of a cube are 1% and 0.25% respectively. The density of the material (ρ) of the cube was determined interms of l and m. The relationship of the fractional errors of the density (ρ), mass (m) and length (l) measurement is given the equation of $\left[\frac{\delta\rho}{\rho}=\frac{3\delta l}{l}+\frac{\delta m}{m}\right]$. The percentage error of the density (ρ) is,

1) 0.50% 2) 0.75% 3) 1.00% 4) 1.75% 5) 3.25%

(13) A passenger who is at rest in a train A which is moving with 10 ms^{-1} velocity towards right respective to the earth observes that another train B, is moving towards him with 60 ms^{-1} velocity along a parallel lane. The velocity of B with respective to A is,

1) 0 ms^{-1} 2) 10 ms^{-1} 3) 50 ms^{-1} 4) 60 ms^{-1} 5) 70 ms^{-1}

(14) 

A fire work on a horizontal floor is explored and a large number of parts are spread every directions in various angles with same u velocity. The maximum area covered by the parts of the fire works on the floor is given by (Neglect the height of the fire work)

1) $\frac{\pi u^2}{g}$ 2) $\frac{\pi u^4}{g^2}$ 3) $\frac{\pi u^2}{g^2}$

4) $\frac{\pi u^2}{g^4}$ 5) $\frac{\pi u^4}{g}$

(15) Ants who are comming out from an ant nest, start their journey at rest with constant 2 cm s^{-2} acceleration. Each and every 2 seconds, one ant is comming out from the nest and they are moving along a linear path. The gap between the second and third ant when the sixth ant is comming out from the nest is,

1) 6 cm 2) 14 cm 3) 21 cm 4) 28cm 5) 36cm

(16) The vehicle 'A' is moving along a horizontal path with 60 km h^{-1} velocity. The vehicle 'B' is moving behind A, with 70 kmh^{-1} uniform velocity. When the gap between these two vehicles is 2.5 km, the vehicle B starts to move with 20 kmh^{-2} retardation. The time taken to passing the vehicle A by the vehicle B is in hours.

1) 0.25 2) 0.5 3) 1 4) 1.5 5) 2

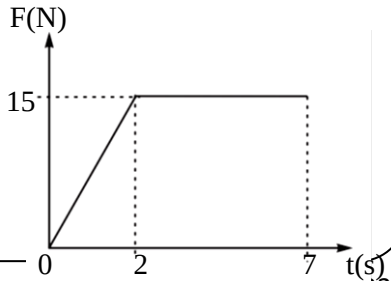
(17) An object is moving 2m distance within it's first 2s and 2.2 m within next 4s. The velocity of the object after the 7th second is

1) 0.1 ms^{-1} 2) 0.2 ms^{-1} 3) 0.3 ms^{-1} 4) 0.4 ms^{-1} 5) 0.5ms^{-1}

(18) A horizontal F force is applied on an object of 3 kg mass, which is kept on a horizontal smooth plane. The variation of the force (F) with time (t) is given in the graph. The velocity of the object after 7 seconds is,

1) 5 ms^{-1} 2) 10ms^{-1} 3) 15ms^{-1}

4) 30 ms^{-1} 5) 70ms^{-1}

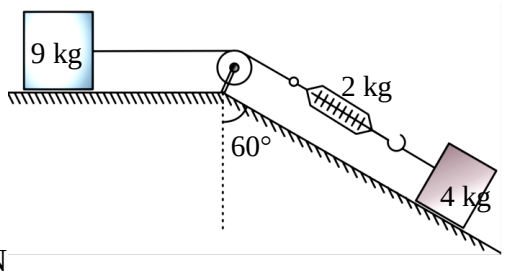


(19) An armored tank which is moving with 30 ms^{-1} velocity towards the north, suddenly changed it's direction to East and moving with 40 ms^{-1} velocity. The magnitude of the thange of the momentum and it's direction is given by,

1) 1 x 10^4 kg ms^{-1} ↖ 2) 1 x 10^4 kg ms^{-1} ↘ 3) 1 x 10^4 kg ms^{-1} ↗

4) 5 x 10^4 kg ms^{-1} ↘ 5) 5 x 10^4 kg ms^{-1} ↗

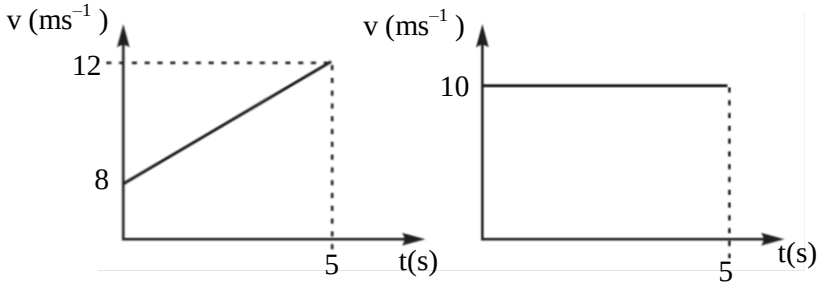
(20) All the surfaces in the figure are smooth. The mass of the spring balance is 2 kg. When the system is smoothly released, the reading of the spring balance is,



1) $6\sqrt{3}$ N 2) 12 N

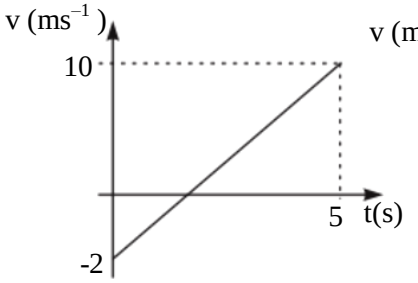
3) $9\sqrt{3}$ N 4) 18 N 5) 20 N

(21) The velocity (v) – time (t) graphs drawn for two athletes A and B, within 5 s time duration at a 100 m event are shown in below,

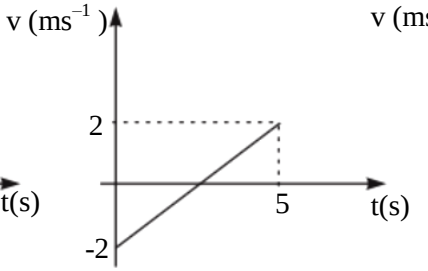


Athlete – A Athlete – B

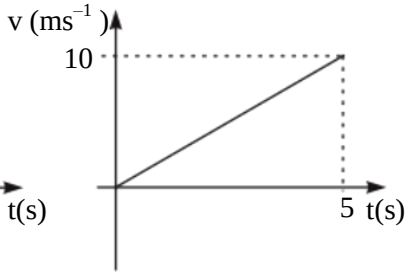
The variation of the velocity of A relative to B with time (t) is best represented by



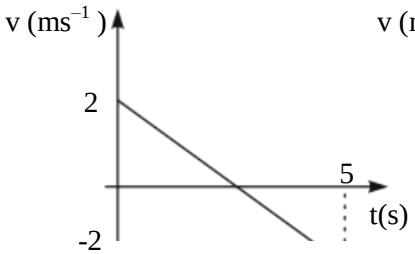
(1)



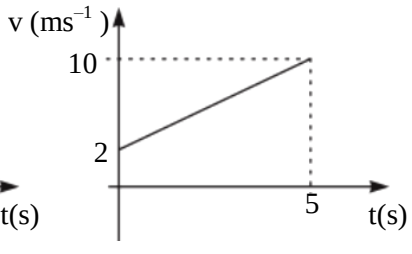
(2)



(3)

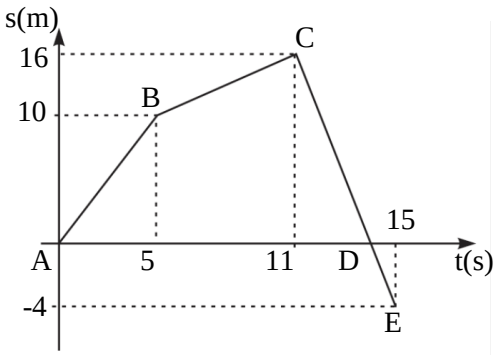


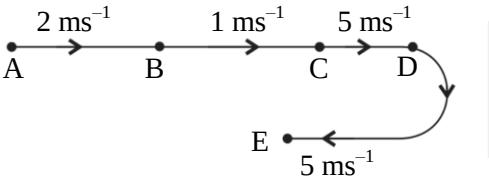
(4)



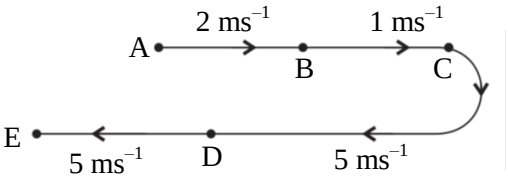
(5)

(22) A small child is walking from point A to point E. The displacement (s) time (t) graph relevent to the motion of the child is given in the figure. The correct path of the motion of the child with correct magnitudes of velocities is given by,

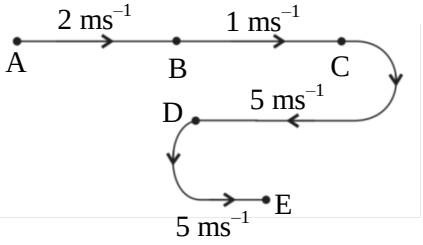




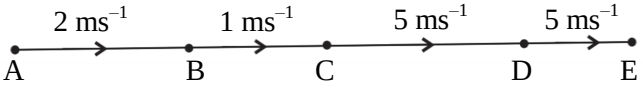
(1)



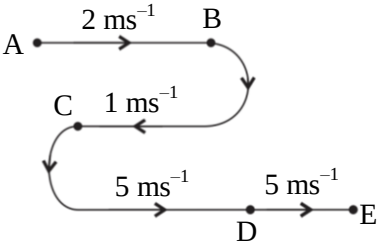
(2)



(3)



(4)



(5)

(23) A horizontal force (P) is given to an object which is kept on a rough horizontal plane. The variation of P with time (t) is given in the graph - 1. The best representation of the variation of the frictional force (F) with time (t) is given by (F_l = Limiting frictional force)

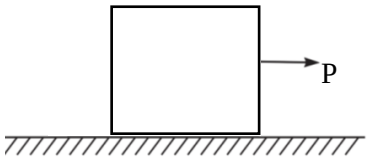
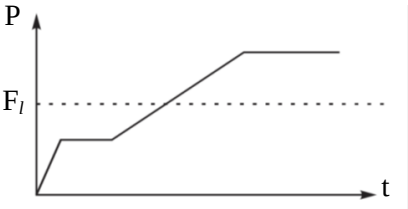
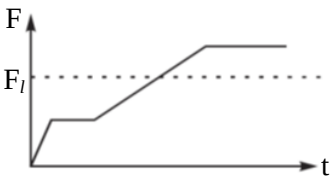


Figure (1)

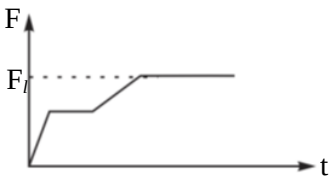


Graph (1)

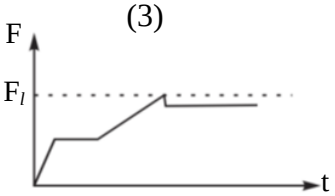
(1)



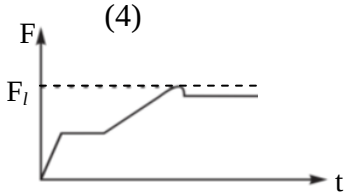
(2)



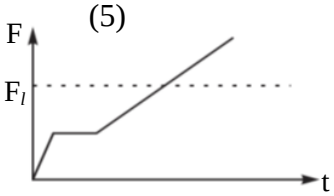
(3)



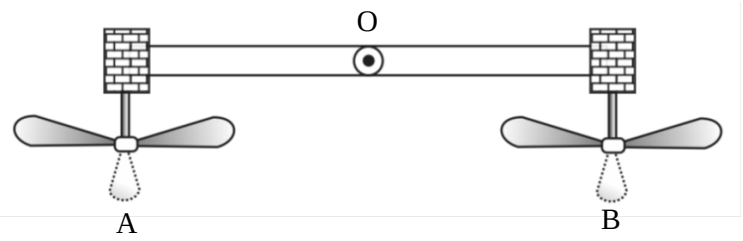
(4)



(5)

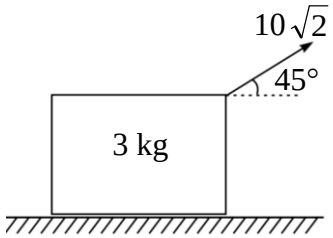


(24) Two identical fans of A and B are connected at the two ends of uniform light rod and it is horizontally balanced as shown in the figure. The system is free to rotate smoothly about the mid point (O) of the rod. The most correct statement, about this system is given by



1) The system is rotating anti-clock wise about the point O, when only the fan A is switched on.
2) The system is rotating clock wise about the point O, when only the fan B is switched on.
3) The system rotates clockwise about the point O, when both A and B are working with same speeds,
4) The system rotates anti -clock wise about the point O, when the fan B is rotating with a higher speed than the fan A.
5) The system rotates anti -clock wise about the point O, when the fan A is rotating with a higher speed than B.

(25) A mass of 3 kg is kept on a rough horizontal plane as shown in the figure. The static and dynamic co-efficient of friction between the mass and the surface are 0.3 and 0.2 respectively. An external force of $10\sqrt{2}$ N, which is 45° inclined to the horizontal is given to the object. The frictional force (F) and the normal reaction (R) are correctly given by,



	(1)	(2)	(3)	(4)	(5)
F(N)	6	6	6	4	4
R(N)	20	30	40	20	30

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Royal College - Colombo 07
Grade 12
First Term test – June 2023
Physics II

01 E II

Time : 3 hours

Important :

- *The question paper consists of **07** pages*
- *The question paper comprises **Part A** and **Part B.** The time allotted for both part is **3 hours***
- *Use of calculators is **not** allowed*

Part A - Structured Essay

(04 pages)

*Answer all the questions on this paper itself**.** Write your answers in this spaces provided is sufficient for your answers and that extensive are not expected.*

Part B - Essay

(02 pages)

*This part contains five questions. Answer only four questions. Use the papers supplied for this purpose. At the end of the time allotted for this paper, tie the two papers so that **Part A** is on top of **Part B** before handing them over to the Supervisor.*

*You are permitted to remove **only** **Part B** of the question paper from Examination hall.*

g = 10 Nkg^{-1}

For Examiner's use only

For the second paper		
Part	Question nos.	Marks
A	1	
	2	
B	3	
	4	
Total		

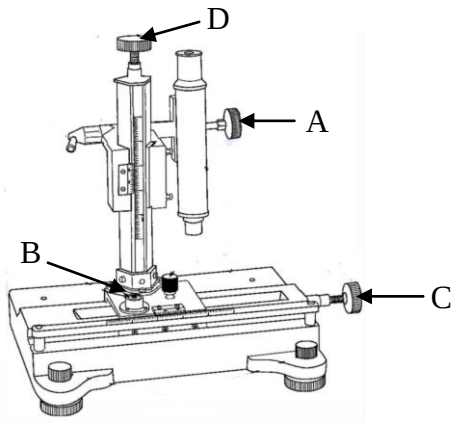
Final Marks

In numbers	
In words	

Part A – Structured Essay

Answer all questions

(1) a) Below figure shows a rough sketch of a travelling microscope which is used in the school laboratory



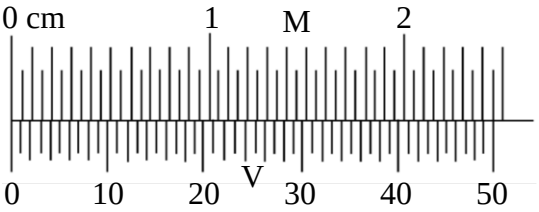
A - ……..…….

B - ……..…….

C - ……..…….

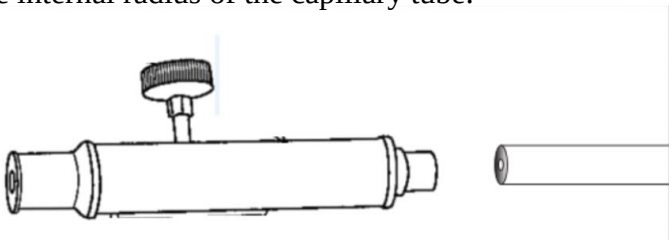
D - ……..…….

b) Below figure shows the positions of the main (M) and the verniner (V) scales of the travelling microscope. What is the least count of this apparatus according to this figure?



Least count = ……..……..……

c) As shown below, the travelling microscope is positioned in front of a capillary tube to determine the internal radius of the capillary tube.



Mark the eye, effective length (focusing length), eye piece and objective in the above figure.

d) If the effective length (focusing length) is not mentioned on the body of the microscope and you have been supplied a square ruled paper, explain briefly how to find the effective length.

…….

…….

…….

e) If the below given readings can be obtained from the travelling microscope put a (✓) and can not be obtained put a (X) in the given brackets.

i) External diameter of a boiling tube ( )

ii) Measurements of a heating object ( )

iii) Height of a virtual image ( )

f) (A) and (B) figures represent the enlarged image of the capillary hole which can be seen though the cross wires of the microscope. It is needed to move the vertical cross wire from X to Y along a vertical diameter of the capillary hole. What is the direction the fine adjustment knob should be rotated for this? Is it clockwise or Anti clockwise?

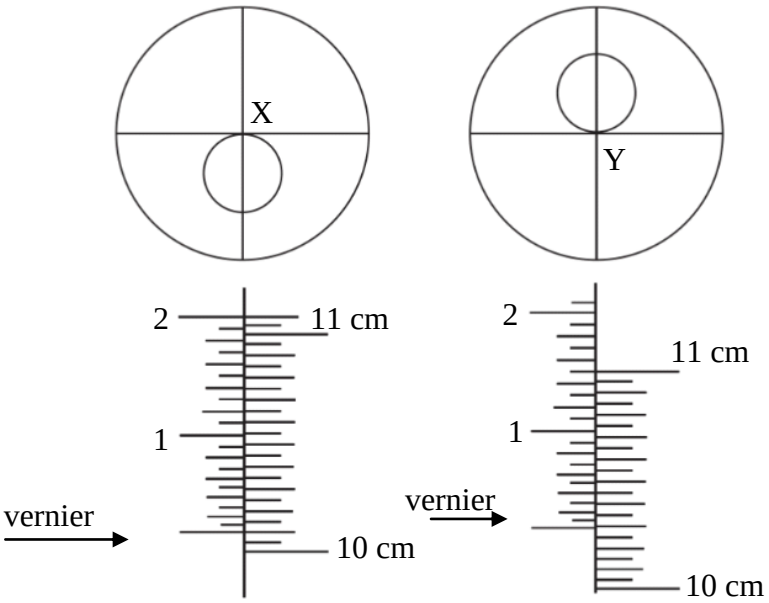


Figure A Figure B

g) What is the need that require two cross wires to be placed inside the microscope?

…….

…….

h) Write down the relevant readings for the above (A) and (B) figures.

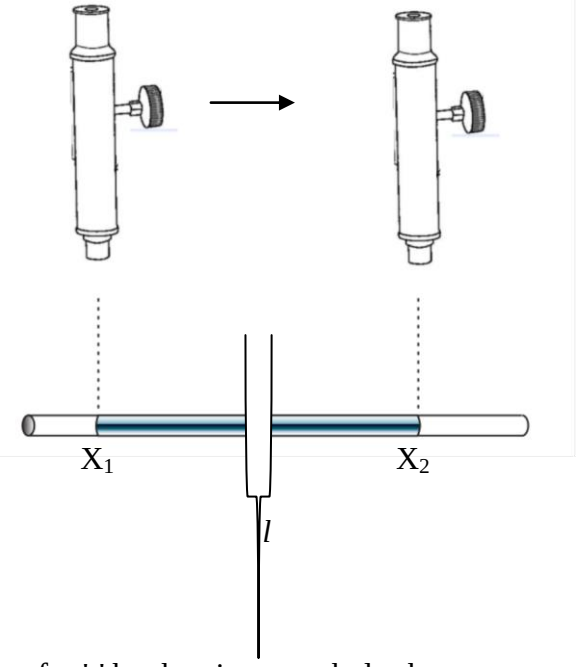
A …… B ……

i) Calculate the diatmeter of the capillary tube according to the above readings.

…….

…….

j) A mercury column is trapped inside the capillary tube to obtain the most accurate mean value for the radius of the capillary tube as shown below. The microscope was adjusted to obtain horizontal scale readings for X_1 and X_2 positions. The distance between X_1 and X_2 is m, the density of mercury is ρ and the radius of the capillary tube is r.



Obtain an expression for 'r' by the given symbols above.

……

……

……

……

(2) You have to determine the mass of an irregular shaped stone using the principle of moment. You have been supplied the stone (mass to be determined), a meter ruler, weights with different values, strings, a pair of scissors and a knife edge.

a) i) What should you do as the initial step of the practical?

……...

……...

ii) What is the reason for this?

……...

b) Explain briefly, how you would practically select the most suitable weight for this practical from the given weights.

…….....

…….....

…….....

c) Draw the practical set up you have used for this practical on the below given knife edge.

d) The mass of the stone and the selected weight are m and m_0 respectively. The distances to the stone and the weight are l and l_0 respectively. Construct a relationship among m, m_0, l and l_0.

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e) There is an idea to determine the mass of the stone using a graphical method by taking the dependent variable as l_0. Rearrange the above expression by showing the independent and dependent variables separately.

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f) Draw the expected graph in the given grid and name the two axis.



g) If the gradient of the above graph is 0.9 and the mass of the weight used is 50g, calculate the mass of the stone.

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h) By using a scale, it has been found that the accurate mass of the stone is 48g. Calculate the percentage error occured due to obtaining the mass of the stone by doing the practical.

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i) Write down two reasons which cause the above error.

1. ……..…

2. ……..…

j) Write down two difficulties you face, if the above practical is done by placing the two weights on the metre rule instead of hanging the weights at the two ends..

1. ……..…

   ……..…

2. ……..…

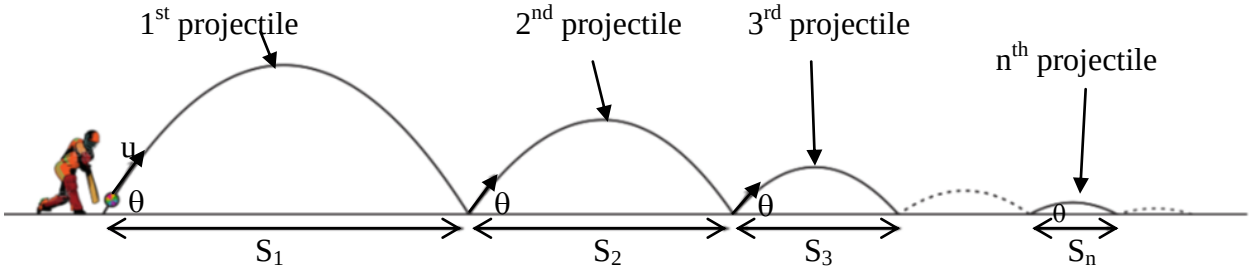
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Part – B – Essay

Answer all questions

(3)



A batsman (A) who plays cricket is ready to bat the ball. He hits the ball, which coming towards the bat, at the ground level. Then the ball is projected with u velocity with θ angle to the horizontal.

a) i) Construct an expression for the horizontal flying time in the first projection.
ii) If the horizontal range for the first projection is S_1,

Show $S_1 = \frac{u^2 \sin(2\theta)}{g}$ [consider $\sin 2\theta = 2 \sin\theta \cos\theta$]

b) Due to the kinetic energy loss of the ball when it hits the ground, the ball bounces with θ angle at a velocity that is half of the velocity which it hits the ground. The ball moves out of the playground after bouncing like this for several times.

i) If the horizontal ranges for the second and third projectiles of the ball are S_2 and S_3, deduce expressions for S_2 and S_3 using the expression obtained for S_1 in (a) (ii).
ii) Show S_1, S_2 and S_3 are in a geometric progression.
iii) What is the initial term of this geometric progression?
iv) What is the common ratio of this progression?
v) Using that, write down the horizontal range (S_n) for nth projection.
[Hint : nth term of a geometric progression is given by ar^{n-1} a is the initial term and r is the common ratio]

c) Draw the displacement-time graph and the velocity-time graph separately for the vertical displacement of the above ball.

d) A batsman hits the ball with 40 ms^{-1} velocity at a 30° angle inclined to the horizontal. In this instance, B fielder who is $175\sqrt{3}$ m from the batsman, starts to run with 10 ms^{-1} velocity towards the batsman to catch the ball.

i) Calculate the horizontal range relevant to the first projectile of the ball.
ii) During the time period relevant to the first projectile of the ball, calculate the displacement of B.
iii) Using that, show that the ball can not be caught by B.

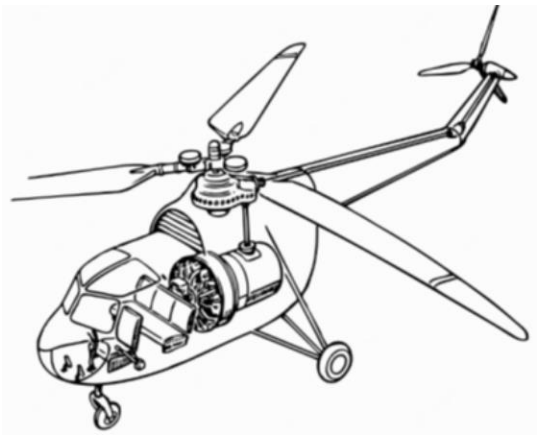
(4) a) i) Write down the Newton's second law.

ii) A motor car starting from rest and uniformly accelerating in 4s, reaches a velocity of $8ms^{-1}$. If the force acting on the car is 2000 N, what is the mass of the car?

iii) During the above time, the constant air resistance acting on the car is 800N and the constant frictional force acting on the car is 200 N. What is the force exerted by the engine?

iv) When the force exerted by the engine becomes zero (when the gear is released), the air resistance and the frictional force mentioned in the above remain constant. What is the distance travelled by the car before it comes to rest?

b) An air craft is retained in space due to an upthrust force created from stream lines. But a helicopter is retained in space due to the rotation of the rotor at a high speed and a stream of air is pushed verticaly downward. When the stream of air is pushed downwards, an upward force is created on the helicopter. Due to that upward force, helicopter travels upwards. The length of a blade of the rotor is 5m, the mass of the helicopter with passengers is 3600kg and the density of air is 1.2 kgm^{-3}. Take $\pi = 3$.



i) Calculate the minimum velocity of the stream of air which should be pushed downwards, to slightly raise the helicopter from the ground.

ii) Calculate the minimum power needed to push the stream of air downwards.

iii) What is the velocity of the air that is pushed downwards when the helicopter is moving in a vertical upward direction with 2.1 ms^{-2} acceleration?

iv) Consider an instance when the helicopter is moving upward with an acceleration but the velocity of the rotor remains constant. When the helicopter is moving upwards, does the acceleration remain constant? Explain.

c) Consider the helicopter is moving forward horizontaly with an acceleration 'a'. An object of mass m is hanging from a string of length l on the roof of the helicopter. Using a and g, write down an expression for the angle made with vertical by the string. g is the gravitational acceleration.

d) Consider an instance where the helicopter is flying horizontaly with a uniform velocity of 10 ms^{-1} at a height of 2 km from the ground. In this instance, an object which is inside the helicopter is gently released. After 10s from releasing, it exploeds into two identical parts and one part drops straight down from the place of the explosion. After the explosion, both parts reached the ground at same time.

i) What is the horizontal velocity of the object after 10s of it being released?

ii) What is the vertical velocity of the object just before the explosion?

iii) Calculate the vertical velocities of the parts after the explosion.

iv) Calculate the horizontal velocities of the parts after the explosion.

v) Calculate the distance between two parts on the ground.